Cross-Sectional Return Prediction in a Small, Illiquid Equity Market: Machine Learning Evidence from the Nairobi Securities Exchange

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Preprint draft generated from reproducible research artifacts

Abstract

This paper asks whether machine-learning models can extract persistent cross-sectional return-ranking information from a small, illiquid emerging equity market, and whether that statistical predictability survives the trading frictions of that market. Using a 12-security Nairobi Securities Exchange universe, walk-forward validation over 2024 provides evidence of positive cross-sectional ranking information. Random Forest is the strongest tested estimator, with mean daily Spearman Information Coefficient (IC) of 0.133 and positive monthly IC in 12 of 12 months; the originally selected LightGBM model achieves mean daily IC of 0.085 in the model-comparison experiment. Feature-group experiments show that simple returns and price information contain much of the useful signal: a returns/price-only ablation achieves mean daily IC of 0.125, compared with 0.111 for the full feature set under the ablation protocol. LLM-derived event variables contribute no measured incremental IC in the evaluated historical window. A retrospective extended out-of-sample evaluation over 2025–2026 remains positive, with mean daily IC of 0.0815 and positive monthly IC in 11 of 15 months, but is not a pristine live holdout. Economically, the evidence is much weaker: an overlapping three-day Top-1 portfolio loses approximately 94.2% cumulatively under a 400 basis point round-trip cost assumption. The main finding is therefore a gap between statistical predictability and implementable profitability in a high-friction market.

1 Introduction

Cross-sectional return prediction is usually studied in large, liquid markets where extensive histories, broad security universes, and low trading costs make statistical signals easier to diversify and monetize. Small emerging markets pose a different problem. They may contain slow-moving information and persistent frictions, but they also offer few securities, intermittent trading, wide effective spreads, limited index data, and high explicit transaction costs. In such markets, evidence of return predictability is not equivalent to evidence of an implementable trading strategy.

This paper studies that distinction using equities listed on the Nairobi Securities Exchange (NSE). The empirical question is: can machine-learning models extract persistent cross-sectional return-ranking information from a small, illiquid emerging equity market, and does that statistical predictability survive the trading frictions of that market? The answer is qualified. Within the studied 12-security universe, the experiments provide evidence of positive out-of-sample ranking information. However, the tested portfolio rules do not establish economic profitability once realistic round-trip costs are imposed.

The paper makes five contributions. First, it provides evidence of persistent cross-sectional ranking information in a small African equity universe. Second, it shows that much of the

predictive information is contained in simple price and return variables rather than in richer feature groups. Third, it documents that estimator choice matters: Random Forest outperforms the originally selected LightGBM model in the 2024 comparison. Fourth, it shows that measurable ranking predictability need not survive trading frictions. Fifth, it provides a reproducible evaluation framework combining walk-forward validation, dependence-aware inference, feature ablations, retrospective extended out-of-sample evaluation, survivorship auditing, permutation importance, and explicit cost-sensitive portfolio simulation.

The remainder of the paper proceeds as follows. Section 2 reviews related work. Section 3 describes the market and data. Section 4 presents the prediction and evaluation methodology. Sections 5–7 report ranking, feature, and economic results. Section 8 summarizes robustness tests, Section 9 discusses limitations, and Section 10 concludes.

2 Related Literature

The study is related to empirical asset pricing and machine learning. Fama and French [4] and a large subsequent literature frame the cross-section of returns as an empirical object explained by security characteristics. Gu et al. [5] show that machine-learning methods can improve expected-return measurement in broad U.S. equity data, especially when nonlinear interactions among characteristics matter. The present paper uses a much smaller and less liquid setting, where the relevant question is not whether a high-dimensional model can improve a large-universe factor portfolio, but whether any rank information survives the sample-size and trading-friction constraints of a small exchange.

The model set includes Random Forests [3] and LightGBM [8]. Both are tree-based estimators, but the empirical results below show that the choice between them matters. The paper is also connected to momentum and reversal evidence, beginning with Jegadeesh and Titman [7]; in the NSE data, simple return and price features are central to the observed predictability.

Illiquidity is central to interpretation. Amihud [1] links illiquidity to expected returns, while Lesmond et al. [9] emphasize the importance of transaction-cost measurement. Emerging equity markets also differ in volatility, integration, and liquidity structure [2]. These considerations motivate the separation between statistical ranking quality and economic profitability.

The inference design responds to known risks in return-predictability research. Overlapping labels and serial dependence motivate HAC/Newey–West standard errors [11]. Multiple-testing and data-mining concerns in the cross-section of expected returns motivate restrained claims and robustness checks [6]. Finally, although recent work studies LLMs as financial text processors or predictors [10], the experiments here do not find an incremental LLM contribution in the evaluated historical window.

3 Market and Data

The NSE is a small, relatively illiquid equity market. The studied universe contains 12 securities: ABSA, BAMB, BAT, BRIT, COOP, EABL, EQTY, JUB, KCB, NMG, SCBK, and SCOM. The main reproduction snapshot spans 2019-01-02 to 2024-12-31 and contains 18,240 security-day rows. The 2024 retrospective out-of-sample evaluation contains 3,048 security-day observations over 254 trading days.

Feature groups include return and price variables, momentum/reversal variables, liquidity and volume proxies, NSE microstructure variables, macroeconomic and exchange-rate variables, and LLM/event variables. The primary implementation target is `label_3d_net_alpha`, a three-day

Table 1: Dataset and evaluation summary.

Quantity	Value
Securities	12
2019–2024 rows	18,240
2019–2024 dates	2019-01-02 to 2024-12-31
2024 test security-days	3,048
2024 test trading days	254
2025–2026 extended rows	3,840
2025–2026 extended trading days	319
Primary target	`label_3d_net_alpha`

forward net-return label after the configured cost assumption. Because historical benchmark data are incomplete, the label should not be interpreted as clean market-model alpha in all periods.

Universe construction is an important limitation. The snapshot is a balanced 12-security seed universe, not a reconstructed historical universe of all NSE listings. Delisted, failed, or historically illiquid securities are absent. Consequently, claims are restricted to the studied universe.

4 Methodology

The predictive task is cross-sectional ranking. For each trading day, models score available securities, and evaluation compares the rank ordering of scores with realized three-day forward net returns. The primary rank metric is the daily cross-sectional Spearman Information Coefficient (IC). Monthly IC values are averages of daily cross-sectional ICs within each calendar month.

Model fitting uses temporal walk-forward validation. For each test month, the training data end before the validation and test windows, and no future test observations are used in fitting. The main 2024 comparison evaluates Random Forest, LightGBM, linear and Ridge regressions, simple reversal and momentum rules, and random rankings. Statistical inference for the daily IC series uses HAC/Newey–West standard errors with lag 3 to account for dependence from overlapping three-day labels.

Economic evaluation uses overlapping three-day sleeves. Each signal date forms a Top-k long-only sleeve using the highest-ranked securities. Sleeves are converted to daily equivalent returns and averaged across active sleeves. The simulation reports cumulative return, annualized volatility, Sharpe ratio, maximum drawdown, sleeve hit rate, and cost sensitivity across round-trip cost assumptions. This design is intentionally simple: it tests whether ranking quality survives a transparent implementation rule before adding portfolio optimization complexity.

5 Empirical Results

5.1 2024 Walk-Forward Ranking Performance

The 2024 LightGBM reproduction gives positive monthly test IC in 11 of 12 months and mean monthly test IC of 0.085. This is evidence of ranking information, but not evidence that LightGBM is the best available estimator. Figure 1 shows monthly IC for selected models.

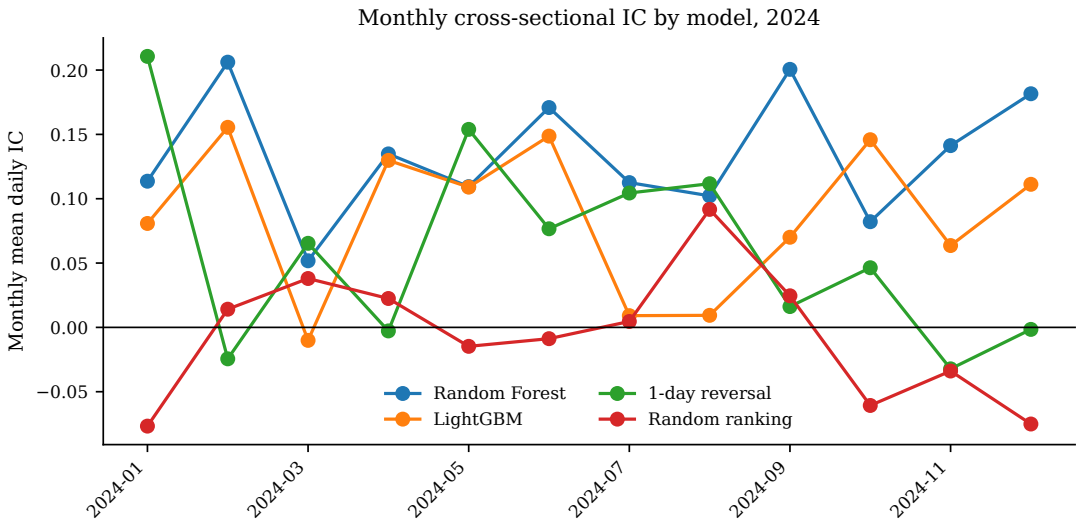


Figure 1: Monthly cross-sectional IC by model. Values are monthly averages of daily cross-sectional Spearman ICs.

Table 2: Model comparison on the 2024 retrospective out-of-sample period.

Model	Mean IC	Median IC	ICIR	Pos. months	Top-1 hit
Random Forest	0.133	0.155	0.411	12/12	16.1%
LightGBM	0.085	0.087	0.279	11/12	16.9%
1-day reversal	0.062	0.042	0.202	8/12	16.5%
3-day reversal	0.056	0.045	0.189	9/12	16.9%
Ridge regression	0.019	0.038	0.060	8/12	16.1%
Linear regression	0.019	0.038	0.060	8/12	16.1%
Random ranking	-0.006	0.000	-0.021	6/12	9.1%
5-day momentum	-0.049	-0.056	-0.157	3/12	14.6%
20-day momentum	-0.052	-0.046	-0.166	4/12	13.4%

5.2 Model Comparison

Table 2 reports the 2024 model comparison. Random Forest has the highest mean daily IC, 0.133, and positive IC in all 12 months. LightGBM achieves mean daily IC of 0.085 and positive IC in 11 of 12 months. Simple reversal rules are positive but weaker, while momentum rules are negative in this horizon and sample.

5.3 Statistical Inference

The LightGBM daily IC series is statistically positive in 2024 under dependence-aware inference. Table 3 reports the HAC estimate. The mean daily IC is 0.0846, with a 95% confidence interval of [0.0437, 0.1254] and p-value 6.59×10^{-5}. This supports a claim of ranking predictability for the evaluated period. It does not establish economic profitability.

5.4 Extended 2025–2026 Retrospective OOS

The extended evaluation covers 2025-01-02 to 2026-03-31. It is best described as retrospective extended out-of-sample evidence, not as a pristine live holdout, because development history would be needed to prove that the period was not inspected during model design. The mean

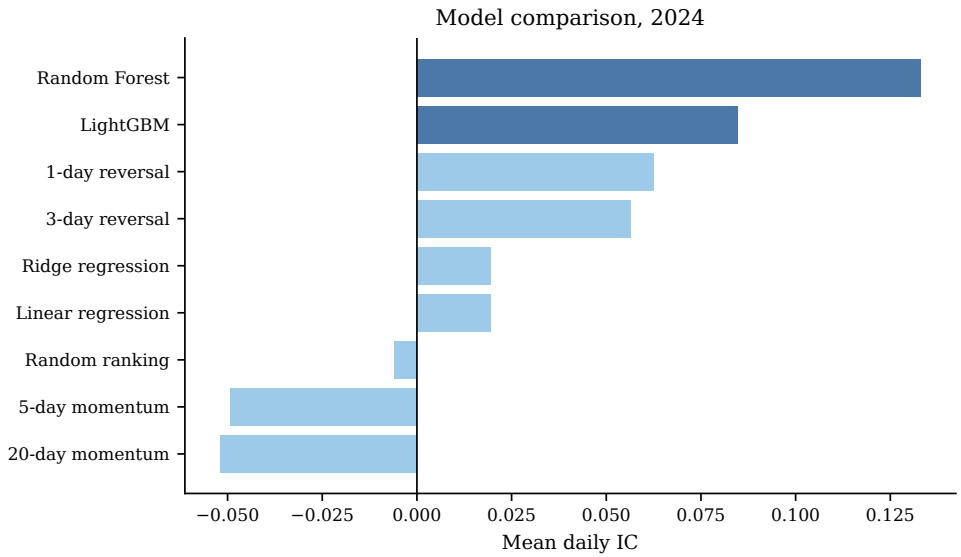


Figure 2: Mean daily IC by model in the 2024 comparison.

Table 3: Dependence-aware inference for the 2024 LightGBM daily IC series.

Statistic	Value
Days	254
Mean daily IC	0.0846
Median daily IC	0.0874
IC standard deviation	0.3027
ICIR	0.2794
HAC lag	3
HAC standard error	0.0208
95% CI	[0.0437, 0.1254]
t-statistic	4.058
p-value	6.59e-05

daily IC is 0.0815 and 11 of 15 months are positive.

6 What Drives Predictability?

Feature-group ablations indicate that simple price and return variables explain much of the predictive performance. In the ablation experiment, the returns/price-only feature set achieves mean daily IC of 0.125, while the full feature set achieves 0.111. This does not imply that additional feature groups are useless in all settings, but it does mean that this experiment does not support a claim that macroeconomic, microstructure, FX, event, or LLM features improve prediction.

Permutation importance is consistent with the ablation evidence. Figure 5 shows that permuting returns/price variables and momentum/reversal variables produces the largest IC drops among interpretable feature groups. LLM/event variables have zero measured IC drop in the historical window. These results should not be read causally: correlated features and small samples can distort both ablation and permutation importance.

Table 4: Extended 2025–2026 retrospective out-of-sample evaluation.

Statistic	Value
Classification	Retrospective extended OOS
Period	2025-01-02 to 2026-03-31
Rows	3,840
Trading days	319
Mean daily IC	0.081
Median daily IC	0.091
Positive months	11/15
Top-1 hit rate	14.4%

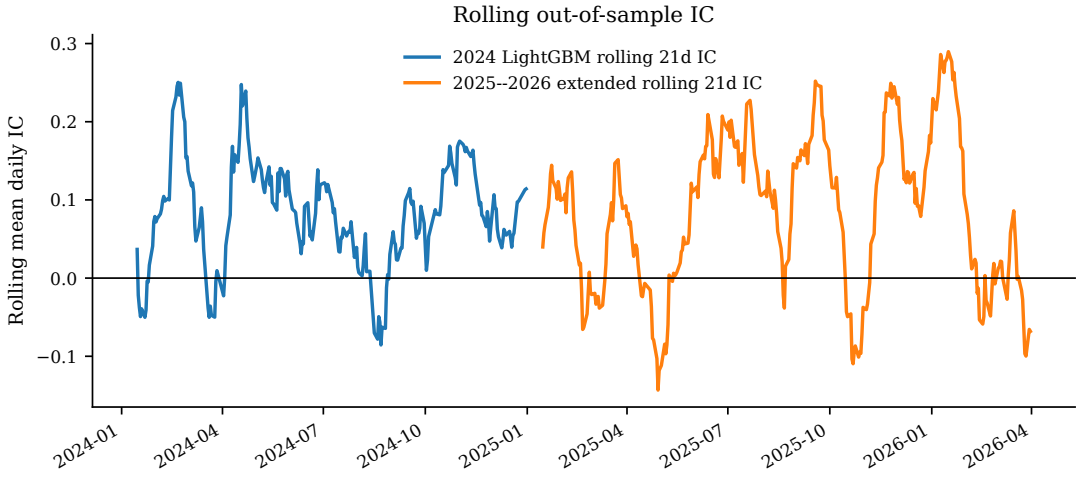


Figure 3: Rolling 21-day out-of-sample IC in the 2024 reproduction and the 2025–2026 retrospective extended evaluation.

7 Economic Evaluation

The portfolio tests show a sharp gap between ranking predictability and economic implementation. Table 6 reports cost sensitivity for overlapping three-day Top-k sleeves. The most important result is the Top-1 portfolio at 400 bps round-trip costs: it loses approximately 94.2% cumulatively in 2024, with maximum drawdown of approximately 94.2%. This result rules out a claim of demonstrated profitability under the tested trading assumptions.

8 Robustness

The completed robustness experiments support the broad existence of ranking information while weakening model-specific claims. Random Forest has mean daily IC of 0.133, positive monthly IC in 12 of 12 months, and outperforms LightGBM in the 2024 comparison. LightGBM results are broadly stable across random seeds in the full-feature ablation protocol, with mean daily IC near 0.109–0.113. A rolling three-year LightGBM window remains positive but weaker, with mean daily IC of 0.083 and positive IC in 9 of 12 months. The one-day and five-day label robustness tests also remain positive, though they answer different economic questions from the primary three-day target.

These checks do not eliminate all concerns. The sample is small, the universe is selected, and the economic tests remain highly sensitive to transaction costs. Robustness here should therefore be

Table 5: Feature-group ablations under the 2024 walk-forward protocol.

Feature set	Mean IC	Median IC	Pos. months	Top-1 hit	Features
Returns/price only	0.125	0.119	12/12	22.0%	9
Full feature set	0.111	0.122	11/12	16.5%	62
Momentum/reversal	0.075	0.063	10/12	19.7%	8
NSE microstructure	0.034	0.040	9/12	13.0%	15
Volume/liquidity	0.002	0.000	5/12	9.4%	9
Macro variables	–	–	–	6.3%	7
KES/USD only	–	–	–	6.3%	2
LLM/event variables	–	–	–	6.3%	10

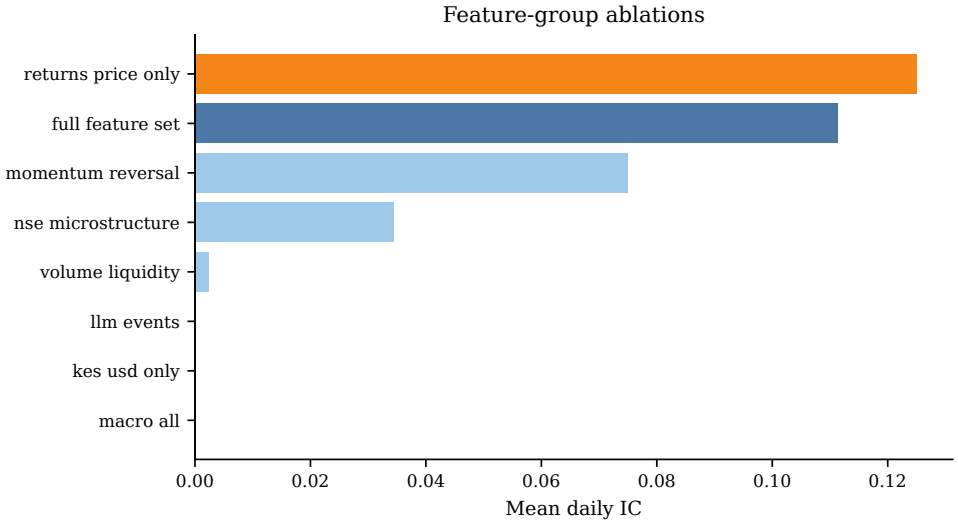


Figure 4: Feature-group ablation results. The returns/price-only feature set outperforms the full feature set in this experiment.

understood as evidence that the ranking result is not a one-off artifact of a single LightGBM seed, not as proof of an implementable strategy.

9 Limitations

Several limitations constrain interpretation. First, the cross-section contains only 12 securities. Security-day row counts therefore overstate the effective sample size because returns share market and macro shocks and labels overlap across dates. Second, the universe is a 12-name seed universe, not a reconstructed historical NSE universe. Survivorship and universe-selection risk remain unresolved. Third, the 2025–2026 experiment is retrospective extended out-of-sample evidence rather than a pristine live holdout. Fourth, transaction costs are estimated rather than observed from order-level execution records. Fifth, full historical index, order-book, and bid–ask data are incomplete, limiting both alpha construction and cost measurement. Sixth, feature-importance results are descriptive rather than causal. Finally, LLM/event variables are not validated as predictive contributors in the historical sample.

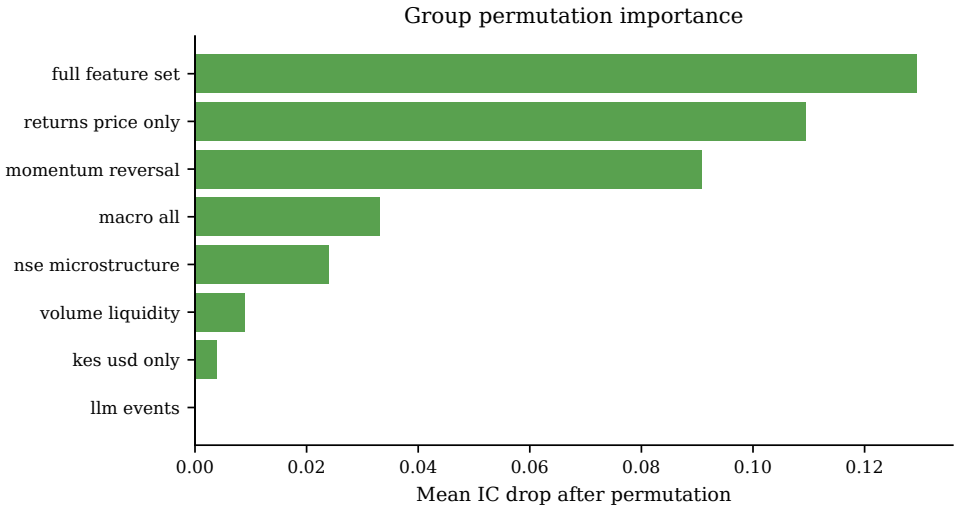


Figure 5: Group permutation importance. Importance is measured as the drop in mean daily IC after permuting each feature group in the test set.

Table 6: Portfolio and transaction-cost sensitivity for overlapping three-day sleeves.

Rule	Cost bps	Total ret.	Sharpe	Max DD	Sleeve hit	Mean sleeve ret.
Top-1	0	80.8%	6.29	-9.9%	57.1%	0.78%
Top-1	100	-22.4%	-1.73	-30.0%	43.3%	-0.22%
Top-1	400	-94.2%	-7.20	-94.2%	16.9%	-3.22%
Top-1	600	-99.0%	-7.47	-99.0%	8.3%	-5.22%
Top-2	0	48.1%	5.06	-5.9%	60.2%	0.51%
Top-2	100	-36.5%	-3.83	-40.0%	40.2%	-0.49%
Top-2	400	-95.3%	-9.83	-95.3%	8.3%	-3.49%
Top-2	600	-99.2%	-10.11	-99.2%	2.8%	-5.49%
Top-3	0	74.3%	8.23	-4.5%	61.0%	0.69%
Top-3	100	-25.2%	-2.78	-32.3%	39.4%	-0.31%
Top-3	400	-94.4%	-10.27	-94.3%	6.7%	-3.31%
Top-3	600	-99.0%	-10.64	-99.0%	3.5%	-5.31%

10 Conclusion

The experiments provide evidence that machine-learning models can extract cross-sectional ranking information from the studied NSE universe. The evidence is strongest for Random Forest in 2024, and the extended 2025–2026 retrospective evaluation remains positive. However, the source of predictability appears simple: returns and price variables explain much of the performance, while richer macro, microstructure, FX, event, and LLM feature groups do not show demonstrated incremental value in the completed experiments.

The economic conclusion is more severe. Under the tested overlapping-sleeve implementation, a Top-1 strategy at 400 bps round-trip costs loses approximately 94.2% cumulatively. The central contribution of the paper is therefore not a profitable trading rule. It is evidence that statistical predictability, ranking quality, economic significance, and implementable profitability are distinct concepts in a high-friction emerging market. In the studied NSE universe, the first two are supported; the last is not.

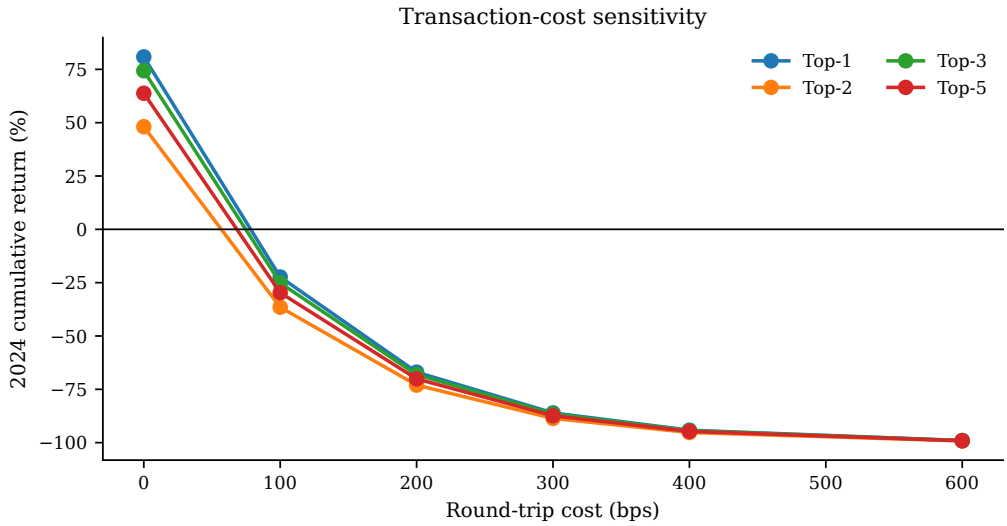


Figure 6: Transaction-cost sensitivity for Top-k overlapping sleeves. The poor 400 bps performance is not hidden by axis truncation.

A Appendix: Full Monthly LightGBM Results

The reproduced 2024 LightGBM monthly test ICs are 0.081, 0.155, -0.010, 0.130, 0.109, 0.149, 0.009, 0.009, 0.070, 0.146, 0.064, and 0.111. The mean monthly test IC is 0.085 and 11 of 12 months are positive.

B Appendix: Implementation Traceability

All main tables and figures are generated by the asset-generation script in the publication directory. The script reads the reproduction, baseline, daily-IC, and phases 9–15 output folders; it then writes LaTeX tables and vector PDF figures into the publication tree. The manuscript consumes only those generated tables and figures.

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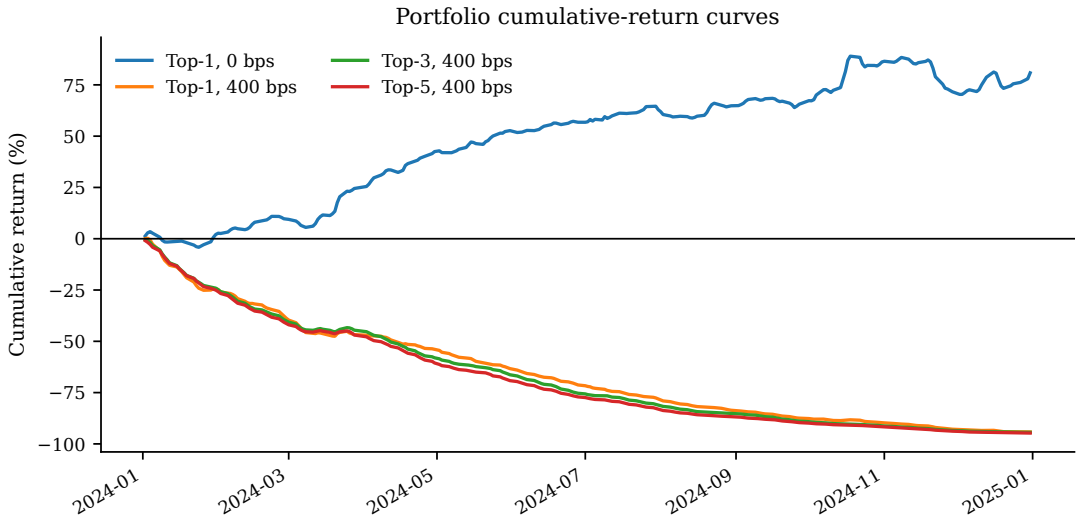


Figure 7: Portfolio cumulative-return curves. The Top-1, 400 bps curve illustrates the economic failure of a naive implementation despite positive ranking IC.

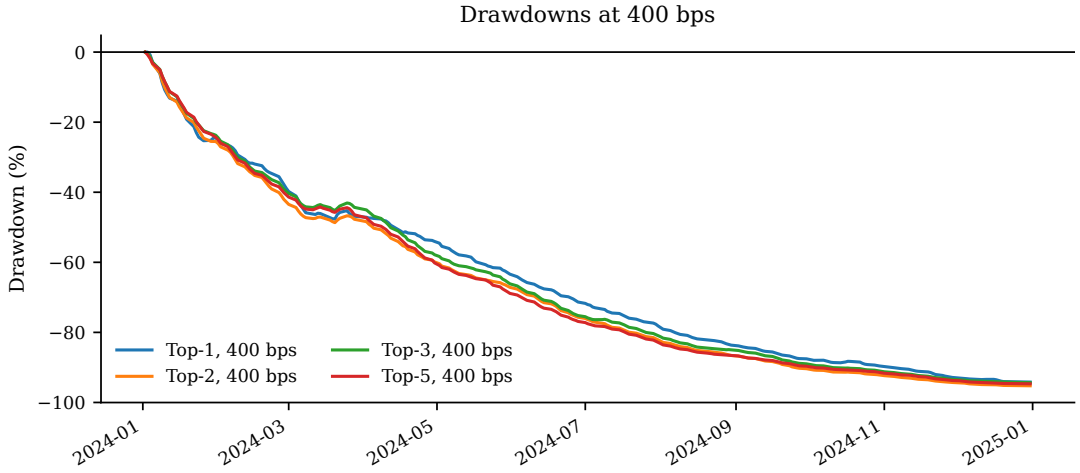


Figure 8: Drawdown curves at 400 bps round-trip costs.

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